Exercises 3

Exercise 1 (Unitary transformations and orthonormal bases)**.** *Let U be a unitary matrix of size $n \times n$ and define vectors $|u_i\rangle = U|i\rangle$, for each $i \in \{1, \ldots, n\}$.*

1. *Show that $|u_1\rangle, \ldots, |u_n\rangle$ is an orthonormal basis of $\mathbb{C}^n$.*
2. *Show that $\sum_{i=1}^n |u_i\rangle\langle u_i| = I$ where $I = \sum_{i=1}^n |i\rangle\langle i|$ is the $n \times n$ identity matrix.*
3. *Show that, for any vector $|\psi\rangle \in \mathbb{C}^n$, $|\psi\rangle = \sum_{i=1}^n \langle u_i|\psi\rangle |u_i\rangle$.*
4. *Let V be another $n \times n$ unitary matrix and let $|v_i\rangle = V|i\rangle$, for each $i \in \{1, \ldots, n\}$. Show that the matrix $W = \sum_{i=1}^n |u_i\rangle\langle v_i|$ is unitary.*
5. *Express W in terms of U and V.*

Exercise 2 (Simulating a measurement in another basis)**.** *Your quantum computer can measure only in the standard basis $|1\rangle, \ldots, |n\rangle$, but your boss gives you a state $|\psi\rangle \in \mathbb{C}^n$ and asks to measure it some arbitrary orthonormal basis $|u_1\rangle, \ldots, |u_n\rangle \in \mathbb{C}^n$. Luckily, you can do this using the following steps:*

(1) apply $U^\dagger = \sum_{i=1}^n |i\rangle\langle u_i|$ to $|\psi\rangle$,

(2) perform the standard basis measurement on the resulting state,

(3) apply U on the post-measurement state.

Verify that these three steps are indeed equivalent to measuring $|\psi\rangle$ in the basis $|u_1\rangle, \ldots, |u_n\rangle$. Namely, check that

1. *the probability to get outcome i in step (2) is $|\langle u_i|\psi\rangle|^2$,*
2. *the final state after step (3) is $|u_i\rangle$.*

Exercise 3 (Basis change)**.** *Express the matrix $X = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ in the basis $\left\{ \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 \\ i \end{smallmatrix}\right), \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 \\ -i \end{smallmatrix}\right) \right\}$.*

Exercise 4 (Entanglement and local operations)**.** *Assume that $|\Psi\rangle$ is an entangled two-qubit state. Show that $|\Psi'\rangle = (U \otimes V)|\Psi\rangle$ is also entangled, for any choice of single-qubit unitary transformations U and V.*

Exercise 5 (Schrödinger equation)**.** *Recall that Schrödinger equation is a differential equation that describes the continuous-time evolution of a quantum system. It looks as follows:*

$$i\frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle,$$

where $|\psi(t)\rangle$ is the state of the system at time t and H is the Hamiltonian that governs the dynamics. The solution of this equation is given by

$$|\psi(t)\rangle = \exp(-iHt)|\psi(0)\rangle$$

where $|\psi(0)\rangle$ is the initial state at time $t = 0$. To understand how such evolution may look like, consider the intial state $|\psi(0)\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and let the Hamiltonian be

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

1. *Compute the state $|\psi(t)\rangle$ at time t.*
2. *Locate this state on the Bloch sphere.*

Exercise 6 (Measuring one qubit at a time)**.** *A two-qubit system is in the following state:*

$$|\psi\rangle = \frac{1}{\sqrt{30}}\big(|00\rangle + 2i|01\rangle - 3|10\rangle - 4i|11\rangle\big).$$

1. *What is the probability of outcome* 1 *if the first qubit is measured in the standard basis?*
2. *What is the state of both qubits afterwards?*
3. *What is the probability that a subsequent standard basis measurement of the second qubit will produce the outcome* 1*?*
4. *What is the probability of obtaining the outcome* 11 *if we would instead measure both qubits of $|\psi\rangle$ simultaneously?*
5. *What is the probability of getting the outcome* 11 *when the two qubits of $|\psi\rangle$ are measured consecutively (first, the first qubit and then the second qubit)? You can use the probabilities derived in steps 1. and 3. to compute this.*